II. GENERAL CONSIDERATIONS

1. TEMPERATURE SCALE

The object in the calibration of any thermocouple is to determine an emf-temperature relation in which the temperature is expressed on a definite and reproducible scale. The International Temperature Scale,[1] adopted in 1927 by thirty-one nations, is now in practically universal use. The methods of realizing this scale are described in detail in the reference cited. The instruments, calibration points, and interpolation equations to be used in the various ranges of the scale are summarized in table 2.

2. GENERAL METHODS

In order to calibrate thermocouples to yield temperatures on the International Scale, it is apparent from the definition that they must be so calibrated that their indications agree with those of the platinum-resistance thermometer in the range −190 to 660° C, the platinum-10 percent rhodium thermocouple in the range 660 to 1,063° C, and the optical pyrometer above 1,063° C. The most direct procedure would therefore be to compare the couples directly with these primary instruments in the appropriate temperature ranges. However, to follow such a procedure in the calibration of every couple requires more time and apparatus than is justifiable or necessary since, in most cases, other methods are available which yield results sufficiently accurate. For example, a thermocouple may be compared indirectly with any of the primary instruments by determining its emf at a number of fixed points, either those which are used in defining the scale or others, the values for which have been determined with the primary instruments. If a few laboratories maintain the apparatus necessary to calibrate thermocouples as working standards to yield temperatures on the International Scale, these standards may be used subsequently to calibrate other couples. This procedure is used far more than any other because the comparison of the indications of two couples is usually simpler than the comparison of two different types of instruments.

The temperature-emf relation of a homogeneous [2] thermocouple is a definite physical property and therefore does not depend upon the details of the apparatus or method employed in determining this relation. Consequently, there are innumerable methods of calibrating thermocouples, the choice of which depends upon the type of couple, temperature range, accuracy required, size of wires, apparatus available, and personal preference.

[1] George K. Burgess. BS J. Research **1**, 635 (1928) RP22.

[2] A homogeneous thermocouple is one in which each element is homogeneous, in both chemical composition and physical condition, throughout its length.